

Australian/New Zealand Standard™

Interior and workplace lighting

Part 2.1: Specific applications— Circulation spaces and other general areas



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AS/NZS 1680.2.1:2008

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Australian Chamber of Commerce and Industry
Australian Electrical and Electronic Manufacturers Association
Energy Efficiency & Conservation Authority of New Zealand
Engineers Australia
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Australian/New Zealand Standard™

Interior and workplace lighting

Part 2.1: Specific applications— Circulation spaces and other general areas

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PREFACE

This Standard was prepared by the Joint Standards Australia/Standards New Zealand Committee LG-001, Interior and Workplace Lighting to supersede AS 1680.2.1—1993, *Interior lighting—Circulation spaces and other general areas*.

This Standard forms Part 2.1 of the AS/NZS 1680 series, which covers lighting of interiors and workplaces. It is intended to be read in conjunction with the general recommendations and requirements of AS/NZS 1680.1:2006 and the structures of the two Standards are closely aligned to facilitate this. It is also noted that, in many areas, parts of various other Standards in the AS/NZS 1680 series will also be applicable, depending on the nature of the building and of the visual tasks involved.

The AS/NZS 1680 series currently consists of the following:

AS/NZS

1680	Interior and workplace lighting
1680.0	Part 0: Safe movement
1680.1	Part 1: General principles and recommendations
1680.2.1	Part 2.1: Specific applications—Circulation spaces and other general areas (this Standard)
1680.2.2	Part 2.2: Specific applications—Office and screen-based tasks
1680.2.3	Part 2.3: Specific applications—Educational and training facilities
1680.2.4	Part 2.4: Industrial tasks and processes
1680.2.5	Part 2.5: Hospital and medical tasks
1680.3	Part 3: Measurement, calculation and presentation of photometric data
1680.4	Part 4: Maintenance of electric lighting systems

NOTE: Until the revision of this series is complete, some of the above Standards might have, as a main title, 'Interior lighting'.

The significant changes that have been made in this Standard in relation to the previous publication include the following:

- (a) Alteration of the minimum uniformity of illuminance from 0.5 to 0.3.
- (b) Expansion of the information and recommendations provided in relation to glare (Section 8).
- (c) Alteration of the recommended maximum glare index in first aid treatment rooms and parking spaces (Appendix D).

The term 'informative' has been used in this Standard to define the application of the appendix to which it applies. An 'informative' appendix is only for information and guidance.

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Australian/New Zealand Standard
Interior and workplace lighting**Part 2.1: Specific applications—Circulation spaces and other general areas**

SECTION 1 SCOPE AND GENERAL

1.1 SCOPE

This Standard sets out recommendations for the lighting of circulation spaces and other areas which commonly occur in buildings of various types. It contains material that adds to or amends the recommendations in AS/NZS 1680.1:2006 and is intended to be read in conjunction with that Standard and with any applicable specific recommendations in other Standards in the AS/NZS 1680 series.

The use of this Standard without reference to AS/NZS 1680.1:2006 is a misapplication of the Standard.

1.2 APPLICATION

If any recommendation in this Standard differs from the general recommendations of AS/NZS 1680.1, the recommendation in this Standard should prevail apart from departures from the illuminance recommendations in this Standard where it should be in accordance with the clauses specifically addressing this matter in Section 3 of AS/NZS 1680.1:2006.

1.3 REFERENCED DOCUMENTS

The following documents are referred to in this Standard:

AS

1735 (all parts) Lifts, escalators and moving walks

AS/NZS

1680 Interior and workplace lighting

1680.1:2006 Part 1: General principles and recommendations

NZS

4332:1997 Non-domestic passenger and goods lifts

1.4 DEFINITIONS

For the purpose of this Standard, the definitions given in AS 1680.1 apply.

SECTION 2 GENERAL REQUIREMENTS OF GOOD INTERIOR LIGHTING

The general requirements of interior lighting should be as recommended in Section 2 of AS/NZS 1680.1:2006.

SECTION 3 TASK VISIBILITY

3.1 GENERAL

The following applies in addition to Section 3 of AS/NZS 1680.1:2006. Where any conflict arises between the two documents, the recommendations in this Section are to be applied.

3.2 RECOMMENDED ILLUMINANCES

For circulation spaces and other general areas, the maintained illuminance should be at least the applicable value given in Appendix D.

3.3 UNIFORMITY OF ILLUMINANCE

The uniformity of illuminance recommendation in Table 3.2 of AS/NZS 1680.1:2006 applies only along the route of travel.

SECTION 4 DIRECTIONAL EFFECTS OF LIGHTING

4.1 GENERAL

The following applies in addition to Section 4 of AS/NZS 1680.1:2006. Where any conflict arises between the two documents, the recommendations in this Section are to be applied.

4.2 REVEALING SIGNIFICANT DETAILS

An important objective of lighting systems for circulation spaces and similar areas is to reveal details of people within the space, including facial expressions and body language. Vertical illuminances within the space are significant for the achievement of this objective and will normally be adequate if—

- (a) the horizontal illuminance at floor level is not less than the value of maintained illuminance recommended in Appendix D (with a uniformity in accordance with Clause 3.3);
- (b) the characteristics of the luminaires are of the type recommended in Clause 9.2; and
- (c) the reflectance and colour of the main interior surfaces comply with the recommendations of Section 6 of AS/NZS 1680.1:2006 and Clause 6.2 of this Standard.

SECTION 5 UNWANTED REFLECTIONS

The recommendations in Section 5 of AS/NZS 1680.1:2006 should apply.

SECTION 6 SURFACES

6.1 GENERAL

The following applies in addition to Section 6 of AS/NZS 1680.1:2006. Where any conflict arises between the two documents, the recommendations in this Section are to be applied.

6.2 REFLECTANCE AND COLOUR OF WALLS AND FLOORS

The finishes used on walls and floors should have a reflectance in accordance with the following:

(a) *Walls*

The wall reflectance should be not less than 0.4 in any instance. Additionally, where the reflectance of the floor is less than 0.2, the wall reflectance should be not less than 0.6.

(b) *Floor*

The reflectance of the floor should be not less than 0.1.

The colours of the above surfaces should have the lowest saturation levels practicable.

SECTION 7 LIGHT SOURCE COLOUR

7.1 GENERAL

The following applies in addition to Section 7 of AS/NZS 1680.1:2006. Where any conflict arises between the two documents, the recommendations in this Section are to be applied.

7.2 LAMP COLOUR APPEARANCE AND COLOUR RENDERING PROPERTIES

The lamps used should be of a type having colour appearance and colour rendering properties complying with the applicable recommendations of Appendix D.

SECTION 8 GLARE AND RELATED EFFECTS

8.1 GENERAL

The following applies in addition to Section 8 of AS/NZS 1680.1:2006. Where any conflict arises between the two documents, the recommendations in this Section are to be applied.

Where the glare evaluation system is used for the assessment of discomfort glare from electric lighting, the calculated glare index should be not greater than the applicable value recommended in Appendix D.

8.2 DISCOMFORT GLARE

Discomfort glare is generally not a problem in circulation areas, as the observer's line of sight is usually a dynamic condition.

Where there is a continually occupied position, e.g. a reception desk or security station, the recommendations on discomfort glare in Section 8 of AS/NZS 1680.1:2006 should be applied.

8.3 DISABILITY GLARE

In circulation spaces located where the general background brightness may be very low, such as galleries, walkways and gangways in power stations or oil refineries, normal discomfort glare recommendations are not relevant.

In these situations the most likely glare condition will be disability glare, in which case the recommendations on discomfort glare in Section 8 of AS/NZS 1680.1:2006 should be applied.

SECTION 9 LIGHT SOURCES, LUMINAIRES AND CONTROL SYSTEMS

9.1 GENERAL

The following applies in addition to Section 9 of AS/NZS 1680.1:2006. Where any conflict arises between the two documents, the recommendations in this Section are to be applied.

9.2 LUMINAIRE CHARACTERISTICS

The luminaire selected should have room surface brightness characteristics which lie within the hatched zone in Figure 9.1.

NOTES:

- 1 The above recommendation relates to the characteristics of the light output distribution of the luminaire which affect its potential to effectively light the surfaces of the interior.
- 2 Room surface brightness characteristics for a variety of basic luminaire types are shown in Section 9 of AS/NZS 1680.1:2006.

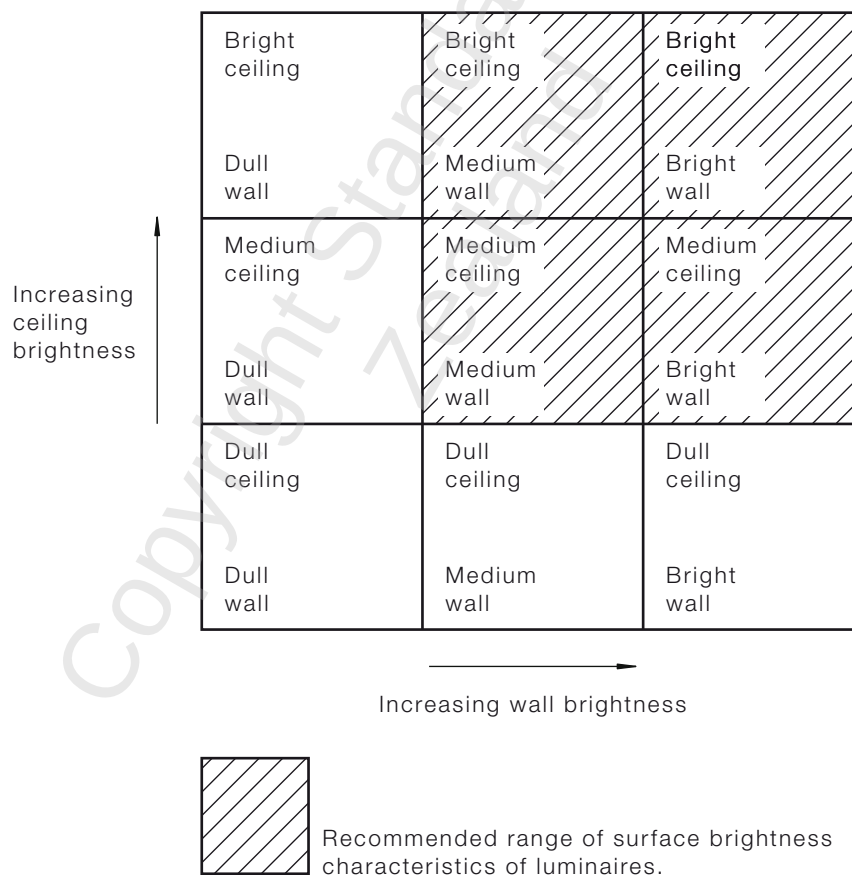


FIGURE 9.1 RECOMMENDED RANGE OF ROOM SURFACE BRIGHTNESS CHARACTERISTICS FOR LUMINAIRES IN CIRCULATION SPACES

SECTION 10 LIGHTING SYSTEMS

Recommendations for lighting systems should be as stated in Section 10 of AS/NZS 1680.1:2006.

SECTION 11 LIGHTING DESIGN PROCEDURE

The lighting design procedure should be as recommended in Section 11 of AS/NZS 1680.1:2006.

SECTION 12 MAINTENANCE OF LIGHTING SYSTEMS AND EQUIPMENT

The maintenance of lighting systems and equipment should be as recommended in Section 12 of AS/NZS 1680.1:2006.

NOTE: Particular attention is to be paid to Clause 12.2 in AS/NZS 1680.1:2006.

APPENDIX A

CALCULATION AND MEASUREMENT OF ILLUMINANCE

(Informative)

The calculation and measurement of illuminance should be as recommended in Appendix B of AS/NZS 1680.1:2006.

APPENDIX B

NOTES ON CHANGES TO ILLUMINANCE RECOMMENDATIONS

(Informative)

Changes to illuminance recommendations are outlined in Appendix C of AS/NZS 1680.1:2006.

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APPENDIX C
NOTES ON THE USE OF UTILIZATION FACTOR TABLES BY THE LIGHTING
DESIGNER

(Informative)

Notes on the use of utilization factor tables by the lighting designer are as recommended in Appendix D of AS/NZS 1680.1:2006.

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APPENDIX D

SPECIFIC RECOMMENDATIONS FOR CIRCULATION SPACES AND OTHER
GENERAL AREAS

(Informative)

D1 GENERAL

Specific recommendations for the lighting of circulation spaces and other general areas of buildings are set out in Table D1. Quantitative recommendations are expressed as minimum values with the exception of the glare index recommendations in Column 5 which are maximum values.

D2 APPLICATION OF TABLE D1

The recommendations of Table D1 should be read in conjunction with AS/NZS 1680.1:2006 and with the following, as applicable:

(a) *Column 1: Type of interior or activity*

If the particular interior/activity is not specifically listed in Column 1, the recommendations for a comparable interior/activity should be adopted, as appropriate.

(b) *Column 2: Maintained illuminance*

See AS/NZS 1680.1:2006 for the definition of maintained illuminance and information on its application in lighting design. The values of maintained illuminance recommended in Column 2 represent the lowest value of the average illuminance over the task area or throughout the interior which should apply at any time during the life of the lighting system. For design purposes, it will be necessary to select an initial illuminance which is significantly greater than the recommended maintained illuminance to allow for the progressive loss of light which will occur over time (see Section 12 and Appendix D of AS/NZS 1680.1:2006).

The values of maintained illuminance recommended in Column 2 should be provided on the appropriate plane in accordance with Section 3 of AS/NZS 1680.1:2006.

(c) *Column 3: Lamp colour appearance group*

The lamp colour appearance groups recommended in Column 3 are those defined in Section 7 of AS/NZS 1680.1:2006 by reference to the correlated colour temperature. Information on the correlated colour temperature of particular lamps is normally given in manufacturers' catalogues.

(d) *Column 4: Lamp colour rendering group*

The lamp colour rendering groups recommended in Column 4 are those defined in Section 7 of AS/NZS 1680.1:2006 by reference to the CIE general colour rendering index, R_a . Information on the CIE colour rendering index of particular lamps is normally given in manufacturers' catalogues.

(e) *Column 5: Maximum glare index*

Section 8 of AS/NZS 1680.1:2006 provides two alternative systems for the control of discomfort glare from electric lighting, namely—

- (i) a glare evaluation system; and
- (ii) a luminaire selection/luminance limiting system.

The recommended values of maximum glare index in Column 5 of Table D1 relate to the glare evaluation system only. Recommendations for the luminaire selection system are fully described in AS/NZS 1680.1:2006 and this system may be used as an alternative to the glare evaluation system for all applications for which a maximum glare index of 19 or more is recommended.

The glare evaluation system is appropriate only for interiors which utilize a regular array of the same type of luminaire.

(f) *Column 6: Other recommendations and advice*

Additional descriptive recommendations and advice are given in Column 6 which are specific to particular tasks or interiors. These recommendations should be taken in conjunction with the quantitative recommendations given in Columns 2 to 5.

TABLE D1
RECOMMENDATIONS FOR CIRCULATION SPACES AND OTHER GENERAL AREAS

	1 Type of interior or activity	2 Maintained illuminance lx	3 Lamp colour appearance group	4 Lamp colour rendering group (minimum)	5 Maximum glare index	6 Other recommendations and advice
1	ENTRANCES					
1.1	Entrance halls, lobbies, foyers	160	1, 2	2	—	The illuminance applies at floor level, See additional recommendations in Section 3 of this Standard. A lower illuminance may be acceptable at night in hotels, restaurants and similar premises where the prevailing illuminances are low (see Clause 1.2). The lighting of vertical surfaces is important to the appearance of the space. Care should be taken with entrance areas to avoid a pronounced change of illuminance between inside and outside, both by day and by night. In addition, the luminaires used should be of such a type or so located that persons entering or leaving the area will not suffer a significant loss of visibility resulting from glare from the luminaires.
1.2	Waiting rooms	160	1, 2	1B, 2	19	The illuminance applies at floor level, See additional recommendations in Section 3 of this Standard.
1.3	Enquiry desks	320	1, 2	2	19	Local lighting may be appropriate.
1.4	Gatehouses	160	1, 2	2	19	Flexible switching or dimming facilities and low surface reflectances may be desirable for security reasons. Special recommendations apply where visual display terminals are used. Care should be taken to avoid light falling on the glazing from directions which will produce reflections that obscure views through the glazing at night. Appropriate exterior lighting giving good vertical illuminances will be necessary for visibility of persons or vehicles passing through the controlled area.
1.5	Loading bays	80	1, 2	2, 3	—	The provision of adequate vertical illuminances at the loading face is important.

(continued)

TABLE D1 (*continued*)

	1	2	3	4	5	6
	Type of interior or activity	Maintained illuminance lx	Lamp colour appearance group	Lamp colour rendering group (minimum)	Maximum glare index	Other recommendations and advice
2	CIRCULATION AREAS					Account should be taken of illuminances in adjacent spaces and pronounced differences should be avoided. See advice given under Item 1.1.
2.1	Corridors, passageways, ramps	40	1, 2	2	—	Vertical illuminances are important for the recognition of people and the detection of obstacles. The illuminance applies at floor level. See additional recommendations in Section 3 of this Standard. A lower illuminance may be acceptable at night in hotels, restaurants and similar premises where the prevailing illuminances are low (see Clause 1.2). The type of luminaires and their location within the space should facilitate the visibility of essential information, e.g. directional or location signs. For heavily trafficked areas, the lighting should generally be provided by luminaires mounted at above head-height to minimize possible obstruction of the light. Special requirements may apply for some corridors, e.g. in hospitals.

(*continued*)

TABLE D1 (continued)

	1	2	3	4	5	6
	Type of interior or activity	Maintained illuminance lx	Lamp colour appearance group	Lamp colour rendering group (minimum)	Maximum glare index	Other recommendations and advice
2.2	Stairs (internal)	80	1, 2	2, 3	—	For uniformity requirements, refer to Table 3.2 of AS/NZS 1680.1:2006. Where the lamps used are of a type which do not restrike immediately from the ‘hot’ condition, consideration should be given to the provision of an alternative source of light (e.g. auxiliary lamps or luminaires) which will give sufficient illumination for safe movement for the period required for the main lamps to restrike. Lamps and bright parts of luminaires should, as far as practicable, be screened from the view of persons ascending or descending the stairs. Lighting should be arranged to provide a contrast between the treads and the risers. Avoid specular reflections on the treads. A single, bold, contrasting stripe set back about 35 mm from the front edge of each tread on a stairway helps to make descent of the stairway quicker and safer in an emergency. This form of striping is more effective than multiple striping, especially in the relatively dim illumination provided by emergency lighting.
2.3	Lift, escalators, moving walks	—	1, 2	2, 3	—	See lighting requirements in the applicable Standard, either in the relevant part of the AS 1735 series or NZS 4332.

(continued)

TABLE D1 (*continued*)

	1	2	3	4	5	6
	Type of interior or activity	Maintained illuminance lx	Lamp colour appearance group	Lamp colour rendering group (minimum)	Maximum glare index	Other recommendations and advice
3	STAFF CANTEENS, CAFETERIA, DINING ROOMS					
3.1	General	160	1, 2	1B, 2	—	Low illuminances may be required where the area is also to be used for special functions. Higher illuminances may be needed for cleaning. Flexible switching or dimming systems should therefore be considered.
3.2	Counters	240	1, 2	1B, 2	—	Attention is drawn to the relevant health and safety regulations.
4	KITCHENS					
4.1	General	160	1, 2	2	22	Attention is drawn to the relevant health and safety regulations. Lamps and luminaires suitable for operation at low temperatures will be necessary for some food storage areas. Lamps and luminaires suitable for hot, humid conditions may be required for other areas.
4.2	Food preparation, cooking, washing up	240	1, 2	1B, 2	22	
5	STAFF ROOMS					
5.1	Changing rooms, locker rooms	80	1, 2	2		Locate luminaires so that there is adequate vertical illuminance on the face of the lockers. Humid conditions may apply in these areas.
5.2	Cleaners' rooms	80	1, 2	2, 3	—	Where the room is to be used for the storage of flammable or corrosive chemicals, the luminaire type and construction should be suitable for the application.
5.3	Cloakrooms	80	1, 2	2	—	Place luminaires outside direct view of resting occupants.
5.4	Rest rooms	40	1, 2	2	—	

(continued)

TABLE D1 (continued)

	1	2	3	4	5	6
	Type of interior or activity	Maintained illuminance lx	Lamp colour appearance group	Lamp colour rendering group (minimum)	Maximum glare index	Other recommendations and advice
6	FIRST AID CENTRES					
6.1	Rest rooms	40	1, 2	1B	—	Place luminaires outside direct view of resting occupants.
6.2	Treatment rooms	320	1, 2	1A	19	Lamp colour rendering is important. Local lighting may be appropriate.
7	TOILETS	80	1, 2	2	—	Where there are mirrors, care should be taken in locating the luminaires to ensure that faces will be adequately lit and to avoid bright reflections of the luminaires from appearing in the mirror.
8	STOREROOMS					
8.1	Rough bulky material:					
	(a) Dead storage	40	1, 2	2, 3	—	
	(b) Live storage	80	1, 2	2, 3	25	
8.2	Medium or fine material requiring care:					
	(a) Dead storage	80	1, 2	2	—	
	(b) Live storage	160	1, 2	2	25	
8.3	Counters:					
	(a) Intermittent tasks	240	1, 2	2	—	
	(b) Sustained tasks	320	1, 2	1B, 2	25	

(continued)

TABLE D1 (continued)

	1	2	3	4	5	6
	Type of interior or activity	Maintained illuminance lx	Lamp colour appearance group	Lamp colour rendering group (minimum)	Maximum glare index	Other recommendations and advice
9	PLANT ROOMS					
9.1	General	80	1, 2	2	25	Luminaires should be positioned having regard to the location of equipment and servicing requirements. Consideration should be given to potential hazards from the lighting producing stroboscopic effects on moving machinery (see Section 9 of AS/NZS 1680.1:2006).
9.2	Control panels, switchboards	160	1,2	1B, 2	—	Recommended illuminance applies in the plane of the control panel or switchboard. Local lighting may be appropriate. Avoid veiling reflections from controls. Too high an illuminance may reduce visibility of signal lights.
10	CONTROL AND MONITORING ROOMS:					Care should be given to minimize unwanted reflections in control and indicating panels and VDU screens.
(a)	Intermittently monitored	240	1, 2	1B, 2	—	
(b)	Continuously monitored	320	1, 2	1B, 2	19	

(continued)

TABLE D1 (continued)

	1	2	3	4	5	6
	Type of interior or activity	Maintained illuminance lx	Lamp colour appearance group	Lamp colour rendering group (minimum)	Maximum glare index	Other recommendations and advice
11	CARPARKS (INDOORS)					The use of light coloured surfaces will improve the interreflection of light within the space and make obstructions (e.g. columns) more visible to drivers.
11.1	Entrances:					
	(a) During daytime					
	—first 15 m	800	1, 2, 3	2, 3	—	Applies over the path of entering vehicles, from the entry point, to facilitate adaptation of drivers from high external illuminances.
	—next 4 m	160	1, 2, 3	2, 3	—	Daylight will normally provide the required illuminance for a distance of approximately twice the height of the entrance.
	(b) During night-time	160	1, 2, 3	2, 3	—	Applies over the path of the entering vehicles, continuing on from the above, to provide a transition for vehicle drivers to the lower illuminances within the interior of the carpark.
11.2	Pay booths	(see recommendations for Gatehouses under Item 1.4)				The enhanced entry lighting recommended in Item (a) should be automatically reduced at night.
11.3	Aisles, ramps, circulating roads, pedestrian crossings	40	1, 2, 3	2, 3	—	
11.4	Normal parking spaces	40	1, 2, 3	2, 3	—	
11.5	Parking spaces for disabled	40	1, 2, 3	2, 3	—	

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